Formula sheet

df = n - 1

$df = min(n_1 - 1, n_2 - 1)$

$s = \sqrt{\frac{\sum_{i=1}^{n}(x_i - \bar{x})^2}{n-1}}$

$\sigma = \sqrt{\frac{\sum_{i=1}^{N}(x_i - \mu)^2}{N}}$

Confidence intervals:

statistic $\pm$ critical value $*$ standard error

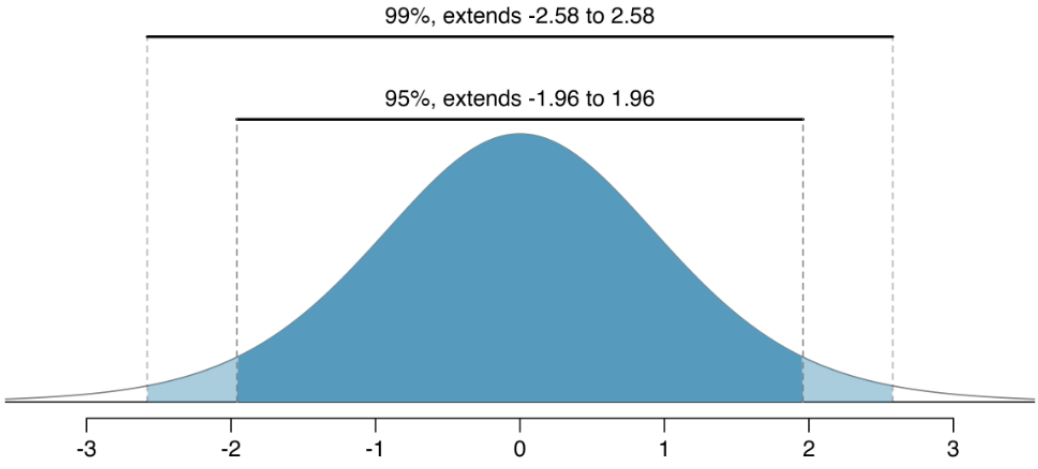
Critical value can be t^* or z^* depending on the distribution we are using

$SE(\hat{p}) = \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$

$SE(\hat{p}_1 - \hat{p}_2) = \sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2}}$

$SE(\hat{p}) = \sqrt{\frac{p_0(1-p_0)}{n}}$

$SE(\hat{p}1 - \hat{p}2) = \sqrt{\frac{\hat{p}pool(1-\hat{p}pool)}{n_1} + \frac{\hat{p}pool(1-\hat{p}pool)}{n_2}}$

$SE(\bar{x}) = \frac{s}{\sqrt{n}}$

$SE(\bar{x}_1 - \bar{x}_2) = \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$